TRIPLE-HASH VERIFICATION PROTOCOL (H■ → H■ → H■)

Digital evidence is highly volatile. The primary challenge in court is proving that a seized electronic device has not been modified between the crime scene and the forensic laboratory.

To satisfy the stringent standards of **Section 63 of the Bharatiya Sakshya Adhiniyam (BSA), 2023**, Malkhana Vault implements the **Triple-Hash Verification Protocol**.

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1. The Three Cryptographic Checkpoints

Malkhana Vault mandates dual-hashing (generating both **SHA-256** and **MD5** simultaneously) at three distinct milestones in the custody lifecycle:

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|                      H1: BIRTH HASH (Crime Scene)                 |
|   - Generated by: Investigating Officer (IO)                      |
|   - Time: Immediately upon device seizure / field acquisition   |
|   - Legal Proof: Proves the original data state at crime scene    |
+--------------------------------+---------------------------------+
                                 |
                                 v  (Transit in Faraday Bag)
                                 |
+--------------------------------+---------------------------------+
|                     H2: RECEIPT HASH (Malkhana Entry)             |
|   - Generated by: Malkhana In-charge / FSL Intake Clerk           |
|   - Time: Before opening the physical seal / placing in drawer    |
|   - Verification: Enforces H1 == H2 (No "Malkhana Gap")           |
+--------------------------------+---------------------------------+
                                 |
                                 v  (Forensic Extraction)
                                 |
+--------------------------------+---------------------------------+
|                    H3: ANALYSIS HASH (Post-Imaging)             |
|   - Generated by: FSL Forensic Examiner                           |
|   - Time: After writing bit-stream image to analysis workstation  |
|   - Verification: Enforces H2 == H3 (Proves no examiner alteration)|
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2. Sequence Diagram

This sequence diagram illustrates how the three hashes are verified and tracked across roles:

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sequenceDiagram
    participant IO as Investigating Officer
    participant APP as Malkhana Vault
    participant DB as SQLCipher DB
    participant MIC as Malkhana In-Charge
    participant FSL as FSL Examiner

    Note over IO,APP: CRIME SCENE — Device Seized
    IO->>APP: NEW_INGESTION → Stage 4
    APP->>APP: dc3dd bit-stream image
    APP->>APP: SHA-256 + MD5 chunked hash
    APP-->>IO: H1 value displayed
    IO->>APP: Confirm + sign (Panch present)
    APP->>DB: evidence.hash_sha256 = H1<br/>evidence.hash_md5 = H1_md5<br/>seized_at = IST timestamp
    APP->>DB: custody_chain → SEIZED entry

    Note over MIC,APP: POLICE STATION — Malkhana Receipt
    MIC->>APP: Scan sealed evidence, enter H1
    APP->>APP: Re-hash original image → H2
    APP->>APP: Compare H1 == H2?
```

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alt H1 == H2 — Integrity confirmed
    APP-->>MIC: ■ MALKHANA ACCEPTED
    APP->>DB: custody_chain → INTAKE<br/>hash_at_transfer = H2<br/>hash_verified = 1
    APP->>DB: archive_matrix slot assigned
else H1 ≠ H2 — Malkhana Gap detected
    APP-->>MIC: ■ HASH MISMATCH — MALKHANA GAP<br/>Evidence may have been accessed in transit
    APP->>DB: custody_chain → INTAKE<br/>hash_verified = 0<br/>notes = MISMATCH FLAGGED
    Note over APP,DB: Certificate engine will<br/>include gap warning
end

Note over FSL,APP: FSL LAB — Analysis
FSL->>APP: Receive sealed device, verify H2
APP->>APP: Re-hash on FSL machine → H3
APP->>APP: Compare H2 == H3?

alt H2 == H3 — No FSL tampering
    APP-->>FSL: ■ ANALYSIS AUTHORISED
    APP->>DB: custody_chain → EXAMINED<br/>hash_at_transfer = H3<br/>hash_verified = 1
else H2 ≠ H3 — FSL alteration detected
    APP-->>FSL: ■ HASH MISMATCH — FSL ALTERATION<br/>Forensic examiner may have modified data
    APP->>DB: custody_chain → EXAMINED<br/>hash_verified = 0<br/>notes = FSL MISMATCH
end

FSL->>APP: Generate Section 63 certificate
APP->>DB: Read all custody_chain entries
APP->>APP: Build certificate with hash audit trail
APP-->>FSL: PDF ready for court submission
```

3. Legal Implications of Mismatches

- **H1 ≠ H2 ("Malkhana Gap"):** Points to tampering or access during transport from the crime scene to the station. Under BNSS Section 153 and BSA Section 63, this mismatch is grounds for the defense to challenge the admissibility of the electronic record.
- **H2 ≠ H3 ("Examiner Alteration"):** Indicates that forensic extraction altered the source evidence (e.g. failure to use a hardware write-blocker).
- **The Vault Safeguard:** Malkhana Vault is an **append-only database**. H1, H2, and H3 values cannot be overwritten or edited once written. All mismatches are highlighted automatically in the generated BSA Section 63 Certificate, ensuring transparent disclosure before the court.